

Tutorial 2

Questions

A rectangular clamped plate has length a in the x -direction and length b in the y -direction. The plate is made of an isotropic homogeneous and linear elastic material and it is subject to a uniform pressure p_0 .

1) Which of the following functions is suitable for applying the Rayleigh-Ritz method to the analysis of the plate described above? Justify your answer.

$$A: w(x, y) = w_0 \sin \frac{\pi x}{a} \sin \frac{\pi y}{b}$$

$$B: w(x, y) = w_0 xy(x - a)(y - b)$$

$$C: w(x, y) = w_0 \sin^2 \frac{\pi x}{a} \sin^2 \frac{\pi y}{b}$$

[20 marks]

2) Having selected an opportune shape function from the list above, compute the expression of the strain energy of the plate.

[20 marks]

3) Calculate the external work performed by the applied pressure.

[20 marks]

4) Determine the transverse displacement at the centre of the plate.

[20 marks]

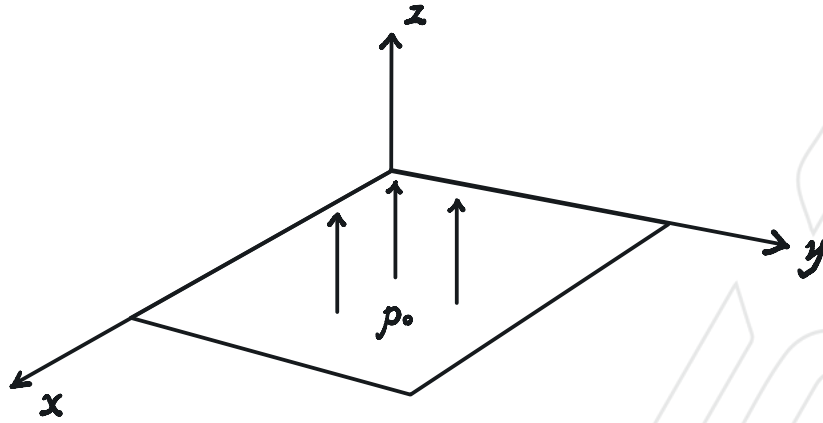
5) Using the approximation from Q2), determine the moments at the centre of the plate assuming that the latter is square, i.e. $b = a$.

[20 marks]

Some useful integrals:

$$\int_0^a \sin \frac{j\pi x}{a} \sin \frac{k\pi x}{a} dx = \int_0^a \cos \frac{j\pi x}{a} \cos \frac{k\pi x}{a} dx = \frac{a}{2} \delta_{jk}, \quad j, k \in \mathbb{N}^+;$$

$$\int_0^a \sin^4 \frac{\pi x}{a} dx = \frac{3}{8} a, \quad \int_0^a \cos \frac{2\pi x}{a} \sin^2 \frac{\pi x}{a} dx = -\frac{a}{4}.$$



1) for clamped plate, essential (geometry) boundary condition should be satisfied, where $w = \frac{\partial w}{\partial n} = 0$ at $(0,0)$, (a,y) , (x,b) and (a,b)

$$A: w|_{\text{edge}} = 0$$

$$\frac{\partial w}{\partial x} = \frac{w_0 \pi}{a} \cos \frac{\pi x}{a} \sin \frac{\pi y}{b} \quad \therefore \frac{\partial w}{\partial x}|_{\text{edge}} \neq 0$$

$$\frac{\partial w}{\partial y} = \frac{w_0 \pi}{b} \sin \frac{\pi x}{a} \cos \frac{\pi y}{b} \quad \therefore \frac{\partial w}{\partial y}|_{\text{edge}} \neq 0$$

$$B: w|_{\text{edge}} = 0$$

$$\frac{\partial w}{\partial x} = w_0 y (x-a)(y-b) - w_0 a x y (y-b) \quad \therefore \frac{\partial w}{\partial x}|_{\text{edge}} \neq 0$$

$$\frac{\partial w}{\partial y} = w_0 x (x-a)(y-b) - w_0 b x y (x-a) \quad \therefore \frac{\partial w}{\partial y}|_{\text{edge}} \neq 0$$

$$C: w|_{\text{edge}} = 0$$

$$\text{for } w = w_0 \sin^2 \frac{\pi x}{a} \sin^2 \frac{\pi y}{b} = w_0 \frac{1}{2} (1 - \cos \frac{2\pi x}{a}) \frac{1}{2} (1 - \cos \frac{2\pi y}{b})$$

$$\frac{dw}{dx} = \frac{w_0 \pi}{a} \sin \frac{2\pi x}{a} \sin^2 \frac{\pi y}{b} \quad \therefore \frac{dw}{dx}|_{\text{edge}} = 0$$

$$\frac{dw}{dy} = \frac{w_0 \pi}{b} \sin^2 \frac{\pi x}{a} \sin \frac{2\pi y}{b} \quad \therefore \frac{dw}{dy}|_{\text{edge}} = 0$$

$$2) U = \frac{1}{2} \iiint \sigma_x \epsilon_x + \sigma_y \epsilon_y + \sigma_z \epsilon_z + \tau_{xy} \delta_{xy} + \tau_{xz} \delta_{xz} + \tau_{yz} \delta_{yz} dx dy dz$$

$$= \frac{1}{2} \iint \int \sigma_x dz \frac{\epsilon_x}{z} + \int \sigma_y dz \frac{\epsilon_y}{z} + \int \tau_{xy} dz \frac{\delta_{xy}}{z} dx dy$$

$$\begin{aligned}
 \therefore U &= \frac{1}{2} \iint M_x \frac{\epsilon_x}{z} + M_y \frac{\epsilon_y}{y} + M_{xy} \frac{\epsilon_{xy}}{z} dx dy \\
 &= \frac{D}{2} \iint - \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right) \left(- \frac{\partial w}{\partial x} \right) - \left(\frac{\partial^2 w}{\partial y^2} + \nu \frac{\partial^2 w}{\partial x^2} \right) \left(- \frac{\partial w}{\partial y} \right) \\
 &\quad - (1-\nu) \frac{\partial^2 w}{\partial x \partial y} \left(- 2 \frac{\partial w}{\partial x \partial y} \right) dx dy \\
 &= \frac{D}{2} \iint \left(\frac{\partial^2 w}{\partial x^2} \right)^2 + \left(\frac{\partial^2 w}{\partial y^2} \right)^2 + 2\nu \left(\frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} \right) + 2(1-\nu) \left(\frac{\partial^2 w}{\partial x \partial y} \right)^2 dx dy
 \end{aligned}$$

where

$$\frac{\partial^2 w}{\partial x^2} = 2w_0 \left(\frac{\pi}{a} \right)^2 \cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{b}$$

$$\frac{\partial^2 w}{\partial y^2} = 2w_0 \left(\frac{\pi}{b} \right)^2 \sin^2 \frac{\pi x}{a} \cos \frac{2\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x \partial y} = w_0 \frac{\pi}{a} \frac{\pi}{b} \sin \frac{2\pi x}{a} \sin \frac{2\pi y}{b}$$

$$\begin{aligned}
 \therefore U &= Dw_0^2 \int_0^a \int_0^b 2 \left(\frac{\pi}{a} \right)^4 \cos^2 \frac{2\pi x}{a} \sin^4 \frac{\pi y}{b} + 2 \left(\frac{\pi}{b} \right)^4 \sin^4 \frac{\pi x}{a} \cos^2 \frac{2\pi y}{b} \\
 &\quad + 4\nu \left(\frac{\pi}{a} \right)^2 \left(\frac{\pi}{b} \right)^2 \sin^2 \frac{\pi x}{a} \cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{b} \cos \frac{2\pi y}{b} \\
 &\quad + (1-\nu) \left(\frac{\pi}{a} \right)^2 \left(\frac{\pi}{b} \right)^2 \sin^2 \frac{2\pi x}{a} \sin^2 \frac{2\pi y}{b} dx dy \\
 &= Dw_0^2 \left[2 \left(\frac{\pi}{a} \right)^4 \int_0^a \cos^2 \frac{2\pi x}{a} dx \int_0^b \sin^4 \frac{\pi y}{b} dy \right. \\
 &\quad + 2 \left(\frac{\pi}{b} \right)^4 \int_0^a \sin^4 \frac{\pi x}{a} dx \int_0^b \cos^2 \frac{2\pi y}{b} dy \\
 &\quad + 4\nu \left(\frac{\pi}{a} \right)^2 \left(\frac{\pi}{b} \right)^2 \int_0^a \frac{1}{2} \left(\cos \frac{2\pi x}{a} - \cos^3 \frac{2\pi x}{a} \right) dx \int_0^b \frac{1}{2} \left(\cos \frac{2\pi y}{b} - \cos^3 \frac{2\pi y}{b} \right) dy \\
 &\quad \left. + (1-\nu) \left(\frac{\pi}{a} \right)^2 \left(\frac{\pi}{b} \right)^2 \int_0^a \sin^2 \frac{2\pi x}{a} dx \int_0^b \sin^2 \frac{2\pi y}{b} dy \right]
 \end{aligned}$$

To solve $\int_0^b \sin^4 \frac{\pi y}{b} dy$,

$$\begin{aligned}
 \int_0^b \sin^4 \frac{\pi y}{b} dy &= \int_0^b \left[\frac{1}{2} (1 - \cos \frac{2\pi y}{b}) \right]^2 dy = \frac{1}{4} \int_0^b 1 - 2\cos \frac{2\pi y}{b} + \cos^2 \frac{2\pi y}{b} dy \\
 &= \frac{1}{4} \left(b - 0 + \frac{b}{2} \right) = \frac{3}{8} b
 \end{aligned}$$

$$\text{similarly, } \int_0^a \sin^4 \frac{\pi x}{a} dx = \frac{3}{8} a$$

$$\therefore U = D\pi^* \left[2 \left(\frac{\pi}{a} \right)^4 \left(\frac{a}{2} \right) \left(\frac{3}{8}b \right) + 2 \left(\frac{\pi}{b} \right)^4 \left(\frac{3}{8}a \right) \left(\frac{b}{2} \right) + \nu \left(\frac{\pi}{a} \right)^2 \left(\frac{\pi}{b} \right)^2 \left(-\frac{a}{2} \right) \left(-\frac{b}{2} \right) + (1-\nu) \left(\frac{\pi}{a} \right)^2 \left(\frac{\pi}{b} \right)^2 \left(\frac{a}{2} \right) \left(\frac{b}{2} \right) \right]$$

$$= D\pi^* \left(\frac{3\pi^4 b}{8a^3} + \frac{3\pi^4 a}{8b^3} + \frac{1}{4ab} \right)$$

$$= \frac{D\pi^4 \pi^*}{4} \left(\frac{3b}{2a^3} + \frac{3a}{2b^3} + \frac{1}{ab} \right) \quad \text{where } D = \frac{Et^3}{12(1-\nu^2)}$$

$$3) W_{op} = \int_0^a \int_0^b q w \, dx \, dy + \oint_A (Q_n w + M_n \frac{\partial w}{\partial n} + M_{nt} \frac{\partial w}{\partial t}) \, ds$$

$$= \int_0^a \int_0^b q w \, dx \, dy$$

$$= \int_0^a \int_0^b p_0 w_0 \sin^2 \frac{\pi x}{a} \sin^2 \frac{\pi y}{b} \, dx \, dy$$

$$= p_0 w_0 \int_0^a \sin^2 \frac{\pi x}{a} \, dx \int_0^b \sin^2 \frac{\pi y}{b} \, dy$$

$$= p_0 w_0 \left(\frac{a}{2} \right) \left(\frac{b}{2} \right)$$

$$= \frac{p_0 w_0 ab}{4}$$

$$4) \Pi = U - W = \frac{D\pi^4 \pi^*}{4} \left(\frac{3b}{2a^3} + \frac{3a}{2b^3} + \frac{1}{ab} \right) - \frac{p_0 w_0 ab}{4}$$

$$\therefore \frac{d\Pi}{d\pi^*} = \frac{D\pi^4 \pi^*}{4} \left(\frac{3b}{2a^3} + \frac{3a}{2b^3} + \frac{1}{ab} \right) - \frac{p_0 ab}{4} = 0$$

$$\therefore w_0 = \frac{p_0 ab}{2D\pi^4 \left(\frac{3b}{2a^3} + \frac{3a}{2b^3} + \frac{1}{ab} \right)}$$

$$= \frac{p_0 a^4 b^4}{D\pi^4 (3b^4 + 3a^4 + 2a^2 b^2)}$$

$$\therefore w = w_0 \sin^2 \left(\frac{\pi x}{a} \right) \sin^2 \left(\frac{\pi y}{b} \right)$$

$$= \frac{p_0 a^4 b^4}{D\pi^4 (3b^4 + 3a^4 + 2a^2 b^2)} \sin^2 \left(\frac{\pi x}{a} \right) \sin^2 \left(\frac{\pi y}{b} \right)$$

$$\therefore w \Big|_{\left(\frac{a}{2}, \frac{b}{2} \right)} = \frac{p_0 a^4 b^4}{D\pi^4 (3b^4 + 3a^4 + 2a^2 b^2)}$$

5) for square plate. $a = b$

$$\therefore \frac{\partial^2 w}{\partial x^2} = 2w_0 \left(\frac{\pi}{a}\right)^2 \cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{a}$$

$$\frac{\partial^2 w}{\partial y^2} = 2w_0 \left(\frac{\pi}{a}\right)^2 \sin^2 \frac{\pi x}{a} \cos \frac{2\pi y}{a}$$

$$\frac{\partial^2 w}{\partial x \partial y} = w_0 \left(\frac{\pi}{a}\right)^2 \sin \frac{2\pi x}{a} \sin \frac{2\pi y}{b}$$

$$\therefore \begin{bmatrix} M_x \\ M_y \\ M_{xy} \end{bmatrix} = D \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & \frac{1-\nu}{2} \end{bmatrix} \begin{bmatrix} -\frac{\partial^2 w}{\partial x^2} \\ -\frac{\partial^2 w}{\partial y^2} \\ -2\frac{\partial^2 w}{\partial x \partial y} \end{bmatrix} \quad \text{where } D = \frac{Et^3}{12(1-\nu^2)}$$

$$\therefore M_x = -2Dw_0 \left(\frac{\pi}{a}\right)^2 \left(\cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{a} + \nu \sin^2 \frac{\pi x}{a} \cos \frac{2\pi y}{a} \right)$$

$$M_y = -2Dw_0 \left(\frac{\pi}{a}\right)^2 \left(\sin^2 \frac{\pi x}{a} \cos \frac{2\pi y}{a} + \nu \cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{a} \right)$$

$$M_{xy} = -D(1-\nu)w_0 \left(\frac{\pi}{a}\right)^2 \sin \frac{2\pi x}{a} \sin \frac{2\pi y}{b}$$

$$\begin{aligned} \therefore M_x \Big|_{\left(\frac{a}{2}, \frac{b}{2}\right)} &= 2D(1+\nu)w_0 \left(\frac{\pi}{a}\right)^2 \\ &= \frac{2(1+\nu)p_0 a^2 a^4}{\pi^2 (3a^4 + 3a^4 + 2a^2 a^2)} \\ &= \frac{2p_0 a^6}{8\pi^2 a^4} (1+\nu) = \frac{p_0 a^2 (1+\nu)}{4\pi^2} \end{aligned}$$

$$\begin{aligned} M_y \Big|_{\left(\frac{a}{2}, \frac{b}{2}\right)} &= 2D(1+\nu)w_0 \left(\frac{\pi}{a}\right)^2 \\ &= \frac{p_0 a^2 (1+\nu)}{4\pi^2} \end{aligned}$$

$$M_{xy} \Big|_{\left(\frac{a}{2}, \frac{b}{2}\right)} = 0$$

Tutorial 2

Solutions

1) The only function that satisfies the geometric boundary condition is **C** (clamped edges have 0 displacement and 0 rotation). Indeed, that is the only function for which: 1) the displacement is zero all along the edges; 2) the derivative of w with respect to x is zero along the edges parallel to the y axis; 3) the derivative of w with respect to y is zero along the edges parallel to the x axis.

2) The strain energy of the plate is given by

$$U = \frac{1}{2} D \int_0^a \int_0^b \left[\left(\frac{\partial^2 w}{\partial x^2} \right)^2 + \left(\frac{\partial^2 w}{\partial y^2} \right)^2 + 2\nu \frac{\partial^2 w}{\partial x^2} \frac{\partial^2 w}{\partial y^2} + 2(1-\nu) \left(\frac{\partial^2 w}{\partial x \partial y} \right)^2 \right] dx dy$$

For the curvatures associated with the assumed shape function C), we find

$$w(x, y) = w_0 \sin^2 \frac{\pi x}{a} \sin^2 \frac{\pi y}{b}$$

$$\frac{\partial w}{\partial x} = w_0 \frac{\pi}{a} \underbrace{2 \sin \frac{\pi x}{a} \cos \frac{\pi x}{a}}_{\sin 2\frac{\pi x}{a}} \sin^2 \frac{\pi y}{b}$$

Recall:

$$\sin 2x = 2 \sin \frac{x}{2} \cos \frac{x}{2}$$

$$\frac{\partial w}{\partial x} = w_0 \frac{\pi}{a} \sin \frac{2\pi x}{a} \sin^2 \frac{\pi y}{b}$$

Similarly:

$$\frac{\partial w}{\partial y} = w_0 \frac{\pi}{b} \sin \frac{2\pi y}{b} \sin^2 \frac{\pi x}{a}$$

$$\frac{\partial^2 w}{\partial x^2} = \frac{2\pi^2 w_0}{a^2} \cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{b}$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{2\pi^2 w_0}{b^2} \sin^2 \frac{\pi x}{a} \cos \frac{2\pi y}{b}$$

$$\frac{\partial^2 w}{\partial x \partial y} = \frac{\pi^2 w_0}{ab} \sin \frac{2\pi x}{a} \sin \frac{2\pi y}{b}$$

The work done by external forces

$$W = \int_0^a \int_0^b p(x, y) w(x, y) dx dy$$

Let us calculate the individual contributions in the expression of the integral giving the strain energy of the plate under bending; we find:

$$U_x = \frac{1}{2} D \int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x^2} \right)^2 dx dy = \frac{2\pi^4 D w_0^2}{a^4} \int_0^a \int_0^b \cos^2 \frac{2\pi x}{a} \sin^4 \frac{\pi y}{b} dx dy =$$

$$= \frac{2\pi^4 D w_0^2}{a^4} \int_0^a \cos^2 \frac{2\pi x}{a} dx \int_0^b \sin^4 \frac{\pi y}{b} dy = \frac{3}{8} \pi^4 D w_0^2 \frac{b}{a^3}$$

And:

$$U_y = \frac{1}{2} D \int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial y^2} \right)^2 dx dy = \frac{2\pi^4 D w_0^2}{b^4} \int_0^a \int_0^b \sin^4 \frac{\pi x}{a} \cos^2 \frac{2\pi y}{b} dx dy =$$

$$= \frac{2\pi^4 D w_0^2}{b^4} \int_0^a \sin^4 \frac{\pi x}{a} dx \int_0^b \cos^2 \frac{2\pi y}{b} dy = \frac{3}{8} \pi^4 D w_0^2 \frac{a}{b^3}$$

And:

$$U_{xy}^{(1)} = D\nu \int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x^2} \right) \left(\frac{\partial^2 w}{\partial y^2} \right) dx dy =$$

$$= \frac{4\pi^4 D \nu w_0^2}{a^2 b^2} \int_0^a \int_0^b \cos \frac{2\pi x}{a} \sin^2 \frac{\pi y}{b} \sin^2 \frac{\pi x}{a} \cos \frac{2\pi y}{b} dx dy =$$

$$= \frac{4\pi^4 D \nu w_0^2}{a^2 b^2} \int_0^a \cos \frac{2\pi x}{a} \sin^2 \frac{\pi x}{a} dx \int_0^b \sin^2 \frac{\pi y}{b} \cos \frac{2\pi y}{b} dy = \frac{\pi^4 D \nu w_0^2}{4ab}$$

And, finally:

$$U_{xy}^{(2)} = D(1-\nu) \int_0^a \int_0^b \left(\frac{\partial^2 w}{\partial x \partial y} \right)^2 dx dy =$$

$$= \frac{\pi^4 D(1-\nu) w_0^2}{a^2 b^2} \int_0^a \int_0^b \sin^2 \frac{2\pi x}{a} \sin^2 \frac{2\pi y}{b} dx dy = \frac{\pi^4 D(1-\nu) w_0^2}{4ab}$$

Adding all the contributions above leads to:

$$U = U_x + U_y + U_{xy}^{(1)} + U_{xy}^{(2)} = \frac{3}{8} \pi^4 D w_0^2 \frac{b}{a^3} + \frac{3}{8} \pi^4 D w_0^2 \frac{a}{b^3} + \frac{\pi^4 D \nu w_0^2}{4ab} + \frac{\pi^4 D(1-\nu) w_0^2}{4ab}$$

$$= \frac{\pi^4 D w_0^2}{8} \left(\frac{3a^4 + 3b^4 + 2a^2 b^2}{a^3 b^3} \right)$$

which gives the expression of the strain energy of the plate.

3) The external work performed by the applied pressure is

$$W = \int_0^a \int_0^b p(x, y) w(x, y) dx dy = p_0 w_0 \int_0^a \int_0^b \sin^2 \frac{\pi x}{a} \sin^2 \frac{\pi y}{b} dx dy = \frac{1}{4} p_0 w_0 ab$$

4) The total potential energy of the plate is

$$\Pi = U - W = \frac{\pi^4 D w_0^2}{8} \left(\frac{3a^4 + 3b^4 + 2a^2 b^2}{a^3 b^3} \right) - \frac{1}{4} p_0 w_0 ab$$

Minimising the total potential energy with respect to the amplitude of the shape function yields

$$\frac{\partial \Pi}{\partial w_0} = 0 \Rightarrow \frac{\pi^4 D w_0}{4} \left(\frac{3a^4 + 3b^4 + 2a^2 b^2}{a^3 b^3} \right) - \frac{1}{4} p_0 ab = 0$$

Thus

$$w_0 = \frac{p_0}{\pi^4 D} \left(\frac{a^4 b^4}{3a^4 + 3b^4 + 2a^2 b^2} \right)$$

which also gives the displacement at the centre of the plate.

5) For a square plate $b = a$; at the centre of the plate ($x_0 = y_0 = a/2$), the curvatures are

$$\frac{\partial^2 w}{\partial x^2} = \frac{2\pi^2 w_0}{a^2} \cos \frac{2\pi x_0}{a} \sin^2 \frac{\pi y_0}{a} = -\frac{2\pi^2 w_0}{a^2}$$

$$\frac{\partial^2 w}{\partial y^2} = \frac{2\pi^2 w_0}{a^2} \sin^2 \frac{\pi x_0}{a} \cos \frac{2\pi y_0}{a} = -\frac{2\pi^2 w_0}{a^2}$$

$$\frac{\partial^2 w}{\partial x \partial y} = \frac{\pi^2 w_0}{ab} \sin \frac{2\pi x_0}{a} \sin \frac{2\pi y_0}{a} = 0$$

Hence, the bending/twisting moments are

$$M_x = -D \left(\frac{\partial^2 w}{\partial x^2} + \nu \frac{\partial^2 w}{\partial y^2} \right), \quad M_y = -D \left(\nu \frac{\partial^2 w}{\partial x^2} + \frac{\partial^2 w}{\partial y^2} \right), \quad M_{xy} = -D(1 - \nu) \frac{\partial^2 w}{\partial x \partial y},$$

$$M_x = M_y = \frac{2\pi^2 w_0}{a^2} (1 + \nu) = \frac{p_0}{\pi^2 D} \left(\frac{2a^2 b^4}{3a^4 + 3b^4 + 2a^2 b^2} \right) (1 + \nu),$$

and

$$M_{xy} = 0.$$